

# Hot Electron Thermoreflectance Coefficient of Gold during Electron–Phonon Nonequilibrium

Elizabeth L. Radue,<sup>†,‡</sup> John A. Tomko,<sup>¶,‡</sup> Ashutosh Giri,<sup>†</sup> Jeffrey L. Braun,<sup>†</sup> Xin Zhou,<sup>§,||</sup> Oleg V. Prezhdo,<sup>||,⊥</sup> Evan L. Runnerstrom,<sup>#</sup> Jon-Paul Maria,<sup>#</sup> and Patrick E. Hopkins<sup>\*,†,¶,○</sup>

<sup>†</sup>Department of Mechanical and Aerospace Engineering, University of Virginia, Charlottesville, Virginia 22904, United States

<sup>¶</sup>Department of Materials Science and Engineering, University of Virginia, Charlottesville, Virginia 22904, United States

<sup>§</sup>College of Environment and Chemical Engineering, Dalian University, Dalian, Liaoning 116622, P. R. China

<sup>||</sup>Department of Chemistry, University of Southern California, Los Angeles, California 90089, United States

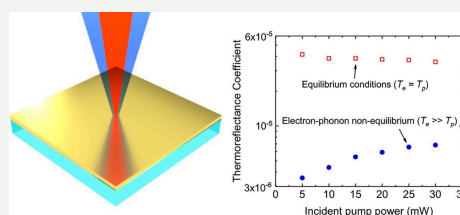
<sup>⊥</sup>Department of Physics and Astronomy, University of Southern California, Los Angeles, California 90089, United States

<sup>#</sup>Department of Materials Science and Engineering, North Carolina State University, Raleigh, North Carolina 27695, United States

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**ABSTRACT:** The temperature-dependent reflectivity of metals is quantified by the thermoreflectance coefficient, which is a material-dependent parameter that depends on the metallic band structure, electron scattering dynamics, and photon wavelength. After short-pulse laser heating, the electronic subsystem in a metal can be driven to temperatures much higher than that of the lattice, which gives rise to unique nonequilibrium electron and phonon scattering dynamics, leading to a “hot electron” thermoreflectance that is different from the traditionally measured equilibrium coefficient. In this work, we analytically quantify and experimentally measure this hot electron thermoreflectance coefficient through ultrafast pump–probe measurements of thin gold films on silica glass and sapphire substrates. We demonstrate the ability to not only quantify the thermoreflectance during electron–phonon nonequilibrium but also validate this coefficient’s predicted dependence on the absolute temperature of the electronic subsystem. The approach outlined in this work provides a metrology to further understand and quantify excited-state scattering effects on the dielectric function of metals.

**KEYWORDS:** thermoreflectance, electron–phonon scattering, pump–probe, electronic temperature, Drude model



Thermoreflectance spectroscopy is a robust technique that has been used to interrogate the band structure and optical transitions in metals for decades.<sup>1–8</sup> In thermoreflectance spectroscopy, an oscillating temperature perturbation is applied to the sample of interest and the AC component of the reflected light is monitored. For example, in a standard optical pump–probe experiment, the pump is a modulated laser beam or train of laser pulses that is used to optically and thermally excite some sample at frequency,  $f$ , while the probe is used to interrogate the small changes in reflectivity of the sample caused by the pump excitation at frequency  $f$ . Due to the

large electronic number density around the Fermi level in metals, this resulting thermorefectivity change is related to characteristics of the electron population that is thermalizing, or thermalized, to a new distribution due to the pump excitation.

Hence, the thermorefectance of the metal, defined as the change in reflectance due to a change in temperature, is given by

$$\frac{\Delta R}{R} = \frac{\Delta R}{R(T_0)} = \frac{R_{\text{excited}} - R(T_0)}{R(T_0)} \quad (1)$$

where  $R$  is the reflectivity,  $T_0$  is the baseline temperature of the sample, and  $R_{\text{excited}}$  is the reflectivity of the metal in the state during and after pump pulse absorption. The thermorefectance of a metal is dictated by its thermorefectance coefficient,  $dR/dT$ , which is typically small, on the order of  $10^{-6}$ – $10^{-4}$   $\text{K}^{-1}$  at room temperature and ambient conditions, its magnitude depending on the metallic band structure, and the probe wavelength.<sup>9–11</sup> Therefore, lock-in techniques are typically used to measure the thermorefectance of metals in pump–probe geometries. Using ultrashort laser pulses in pump–probe geometries, the time evolution of the thermorefectance can be measured.<sup>12,13</sup>

In eq 1, the parameter containing all of the pertinent electronic and atomistic information is  $R_{\text{excited}}$ , which is the reflectivity of the probe pulse after interacting with the metal in the excited state from the pump pulse. Upon pump pulse absorption, the electrons in the metal redistribute themselves to different electronic states depending on the energy of the photons absorbed by the metal; therefore,  $R_{\text{excited}} = R(T_e, T_p)$  during pump absorption is related to the electronic band structure and available states for electron redistribution.<sup>7,14–17</sup> Upon and during thermalization (or partial thermalization),<sup>18,19</sup> the electron system can be described by an evolving electron temperature,  $T_e$ , which is greater than  $T_0$  and different from the phonon temperature,  $T_p$ ; this excited electron temperature and its evolution and decay is governed by the power of the pump pulse along with the electron dynamics and scattering mechanisms in the metal.<sup>14,20–33</sup>

Thus, monitoring the time evolution of the magnitude of the probe thermorefectance signal gives an indirect indication of the electronic states and scattering dynamics in metals. This relationship underlies the principle of thermorefectance-based measurement techniques, as the changes in electronic energies are related to the temperature evolution of the system. While the thermorefectance responses of various materials have been studied for quite some time,<sup>34–36</sup> direct experimental measurements of the thermorefectance coefficients of materials using the short-pulsed measurement platforms employed in ultrafast and nanoscale energy transport measurements (e.g., time domain thermorefectance (TDTR) and frequency domain thermorefectance (FDTR))<sup>37–41</sup> have only become absolutely quantifiable recently.<sup>9,10,42–44</sup> However, these values for  $dR/dT$  using TDTR are measured during a temporal regime when the electrons and phonons have equilibrated and can be described with very similar temperatures (i.e.,  $T_e \approx T_p$ ). In other words,  $dR/dT$  is measured at pump–probe time delays hundreds of picoseconds after pulse absorption, orders of magnitude longer than the actual absorption event. For example, in a metal film, electron–phonon equilibrium occurs in the first several picoseconds after subpicosecond pulse absorption, and  $dR/dT$  is typically measured at time delays on the order of hundreds of picoseconds. Since the reflectivity of materials depends on both the electronic and lattice temperatures ( $R(T_e, T_p)$ ), the quantification of  $dR/dT_e$  independently of  $dR/dT_p$  would provide a unique avenue to study the electronic response of materials after short pulsed laser heating and during conditions of electron nonequilibrium.

In this work, we quantify the “hot electron” thermorefectance coefficient in gold films via monitoring the thermorefectance of the gold during pump–probe TDTR measurements immediately after pulsed laser absorption. During this regime, the electrons in the Au are out of equilibrium with the gold lattice, and thus, the condition  $T_e \approx T_p$ , usually assumed while reporting  $dR/dT$  values, is invalid. In this pump–probe time delay measurement regime, which is on the order of the subpicosecond pulse duration,  $T_e$  can be much greater than  $T_p$ , allowing us to measure this hot electron thermorefectance coefficient,  $dR/dT_e$ .

Using this approach applied to thin gold films on silica glass and sapphire substrates as experimental examples, we validate the linear electron temperature dependence of the thermorefectance coefficient. Furthermore, we demonstrate a lattice temperature dependence in  $dR/dT_e$ . Traditionally, the nonlinear reflectivity change in free electron metals has only been associated with changes in the electronic temperatures, and our lattice temperature dependence of the measured  $dR/dT_e$  suggests this is not a complete picture. Indeed, as we derive below, an apparent nonlinear temperature dependence could arise from a consequence of additional heating from the absorbed laser inducing a steady-state temperature rise in thermally insulating substrates. This increase in lattice temperature in our TDTR experiments, driven by pulse accumulation,<sup>45,46</sup> has been a well established artifact of TDTR experiments on low thermal conductivity substrates, and our previous work has studied this effect on electron–phonon relaxation after short pulsed laser heating.<sup>47</sup> Our approach outlined in this work to directly quantify  $dR/dT_e$  allows us to study this effect on the ultrafast optical response directly, along with providing a metrology to further understanding the excited-state scattering effects on the dielectric function of materials.

## ■ THEORY OF TEMPERATURE DEPENDENCE OF $dR/dT_e$

Quantitatively, any thermorefectance signal can be mathematically represented by a change in the baseline reflectivity of a sample surface resulting from a change in temperature of the sample (eq 1). The reflectivity of a material at the air (vacuum)/film interface is given by

$$R = \frac{(n_1 - 1)^2 + n_2^2}{(n_1 + 1)^2 + n_2^2} = \left| \frac{1 - \sqrt{\tilde{\epsilon}}}{1 + \sqrt{\tilde{\epsilon}}} \right|^2 \quad (2)$$

where  $n_1$  and  $n_2$  are the real (refractive index) and imaginary (extinction coefficient) parts of the complex index of refraction,  $\tilde{n}$ , and  $\tilde{\epsilon}$  is the complex dielectric function. For an optical excitation that only promotes intraband transitions, such as free electron metals in the near-infrared regime, we assume a Drude model for  $\tilde{\epsilon}$ , given by

$$\tilde{\epsilon}_f = 1 - \frac{\omega_{\text{plasma}}^2}{\omega \left( \omega + \frac{i}{\tau} \right)} \quad (3)$$

where  $\omega$  is the frequency of light (i.e., 800 nm for this work),  $\omega_{\text{plasma}}$  is the plasma frequency, and  $\tau$  is the scattering time of the carriers interacting with the incoming light at frequency  $\omega$ . Note, for the experimental measurements in this work on gold in which the thermorefectivity is monitored with a 800 nm pulse, the use of a Drude model to approximate  $\tilde{\epsilon}$  is well established.<sup>11,22,48–50</sup>

The temperature dependence of the dielectric function, and thus temperature dependence of optical reflectivity, for metals is rooted in the temperature dependence of the electronic scattering rate,  $\tau^{-1}$ . This relaxation time is dependent on both electron–electron (e–e) and electron–phonon (e–p) scattering rates and can be expressed as  $\tau^{-1} = \tau_{e-e}^{-1} + \tau_{e-p}^{-1}$ . For the free electron metal focus in this work, we make the common assumption that  $\tau_{ee}^{-1} = A_{ee}T_e^2$  and  $\tau_{ep}^{-1} = B_{ep}T_p$ , where  $A_{ee}$  and  $B_{ep}$  are constants relating to electron–electron and electron–phonon scattering processes.<sup>51</sup> Note, the plasma frequency,  $\omega_p$ , can also vary with respect to temperature and affect the dielectric function/optical reflectivity; as discussed in-depth in the Supporting Information, this weak temperature dependence does not contribute significantly to the laser parameters used in this work and can thus be considered constant with respect to temperature in our following derivations. Thus, taking the derivative of eq 2 with respect to  $T_e$  yields  $dR/dT_e$ . While this derivative is trivial in practice, especially with the use of any commercially available computing software (e.g., Mathematica), the analytical result is quite tedious. Thus, it is more intuitive to consider the series expansion of this result. Additionally, expansion about the absolute temperature may be more representative of values measured in such cases of nonequilibria; as stated via eq 1, pump–probe experiments measure the change in reflectivity due to pump excitation, not necessarily the differential, and are thus sensitive to absolute temperature at a given time delay rather than the fractional change in temperature. Expanding  $dR/dT_e$  about  $T_e$  and  $T_p$  yields

$$\frac{dR}{dT_e} \approx \sqrt{R}(\tilde{C} + \tilde{D}T_p)T_e \quad (4)$$

where  $\tilde{C}$  and  $\tilde{D}$  are both complex numbers that are functions of  $\omega$ ,  $\omega_{\text{plasma}}$ ,  $A_{ee}$ , and  $B_{ep}$  (note,  $\tilde{C}$  and  $\tilde{D}$  are presented in the Supporting Information). From eq 4, we see a trend that is in contrast with the typical assumption that the origin of a nonlinear reflectivity is only proportional to  $T_e^2$ . More specifically, when  $T_p$  is non-negligible relative to  $T_e$ , nonlinear dependence in the change in reflectivity would be influenced by  $T_p$ , and thermorefectance coefficient takes the form

$$\frac{dR}{dT_e} \approx \tilde{E}T_e + \tilde{F}T_pT_e \quad (5)$$

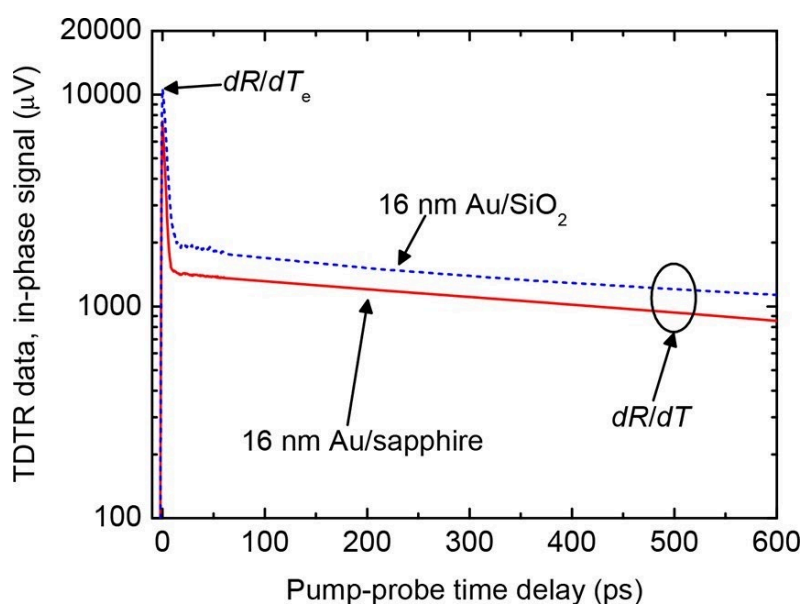
However, from eq 5, when  $T_p \ll T_e$ ,

$$\frac{dR}{dT_e} \approx \tilde{G}T_e \quad (6)$$

where  $\tilde{E}$ ,  $\tilde{F}$ , and  $\tilde{G}$  are functions of  $\omega$ ,  $\omega_p$ ,  $A_{ee}$ ,  $B_{ep}$ , and  $\sqrt{R}$ . Equation 6 implies that, when  $T_p \ll T_e$ , the hot electron thermorefectance coefficient should appear linear and lattice temperature independent, and thus,  $\Delta R \propto T_e^2$ . However, when  $T_p$  is large, additional nonlinearities in  $\Delta R$  could arise from lattice heating. Indeed, several previous reports of this nonlinear change in reflectivity were observed in thin Au films on amorphous SiO<sub>2</sub> substrates,<sup>11,49,52</sup> where the average absorbed power from the high repetition rate laser sources can lead to pulse-to-pulse accumulation of absorbed energy resulting in an appreciable steady-state temperature rise, leading to phonon temperature rises well above ambient conditions.<sup>47</sup> Without explicit measurements of  $dR/dT_e$ , which we demonstrate in this work, the true behavior of  $\Delta R$  with  $T_e$  would be left unverified experimentally. Our measurements of  $dR/dT_e$ , as described below, allow us to rectify this discrepancy in the nonequilibrium free electron response of materials.

## ■ EXPERIMENTAL MEASUREMENT OF $dR/dT_e$

The gold films studied in this work were 16 nm thick as confirmed with picosecond acoustics; the details of this analysis can be found in the Supporting Information.<sup>53</sup> The films were electron-beam evaporated on amorphous SiO<sub>2</sub> and (0001) oriented single-crystal sapphire substrates after a light alcohol rinse and O<sub>2</sub> plasma clean to each substrate. The hot electron thermoreflectance coefficient was measured using a standard pump-probe TDTR configuration centered around a Ti:Al<sub>2</sub>O<sub>3</sub> oscillator with a 80 MHz repetition rate, center wavelength of 800 nm, and bandwidth of 10.5 nm.<sup>37–41</sup> The pump path is focused into a BiB<sub>3</sub>O<sub>6</sub> crystal,<sup>54</sup> converting the pump to a 400 nm central wavelength. The probe path is directed down a mechanical delay stage, and the path length is varied to alter the temporal delay between the arrival of the probe beam relative to the pump beam on the sample surface. Thus, our measurements of  $dR/dT_e$  (and  $dR/dT$ ) in the thin Au films are collected at 800 nm. The cross-correlation of the pump and probe pulses at the sample surface is measured to be ~825 fs, as outlined in the Supporting Information. Example TDTR data on the Au/sapphire and Au/SiO<sub>2</sub> samples are shown in Figure 1, along with indicators of the pump-probe delay time in which  $dR/dT_e$  and  $dR/dT$  are measured.



**Figure 1.** TDTR data for the 16 nm thin films on SiO<sub>2</sub> and Al<sub>2</sub>O<sub>3</sub> substrates. At short time scales, the large rise in the measured change in reflectivity is associated with an electron–phonon nonequilibrium when  $T_e \gg T_p$ . By ~10 ps, the electrons and phonons in gold film are thermalized and  $T_e \approx T_p$  for  $t > 10$  ps.

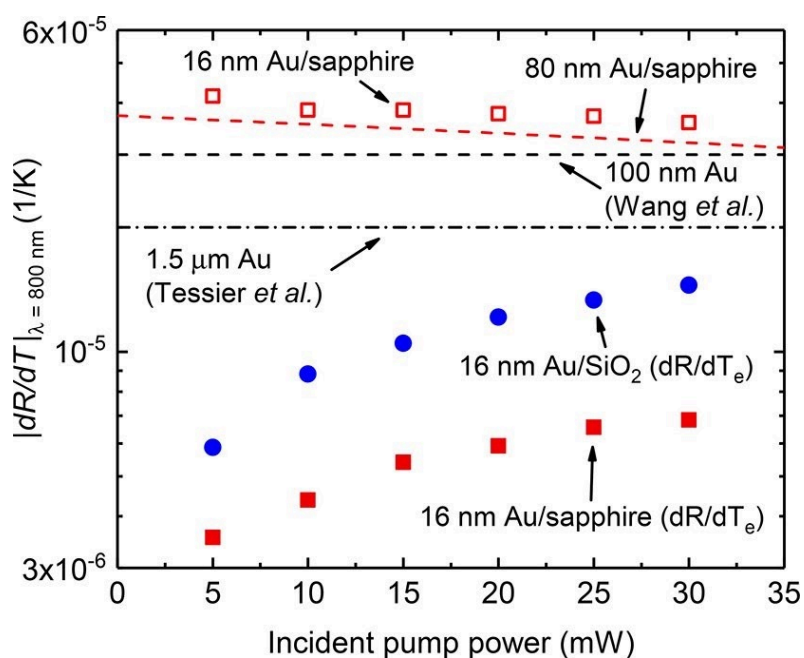
We quantify  $dR/dT_e$  (and  $dR/dT$ ) using the procedure outlined in Wang et al.<sup>10</sup> In this approach,  $dR/dT$  is determined by

$$\frac{dR}{dT} = \frac{\sqrt{2}}{Q} \left[ \frac{V(t)}{V_0} \right] \left[ \frac{R}{\Delta T(t)} \right] \quad (7)$$

where  $Q$  is the quality factor of the LC circuit used to filter out side bands (in our system  $Q = 10$ ),  $V_0$  is the DC voltage measured by the photodiode,  $V(t)$  is the change in voltage measured by the lock-in amplifier at some time pump-probe delay time  $t$ ,  $R$  is the reflectivity of the Au film in equilibrium, and  $\Delta T(t)$  is the change in temperature of the Au film induced by the absorbed pump pulse at some pump-probe delay time  $t$ . In the case of the equilibrium thermoreflectance

coefficient  $dR/dT$ , which is commonly reported when  $T_e \approx T_p$ ,  $\Delta T(t)$  is determined by calculating the surface temperature predicted via the solution to the cylindrical heat equation detailed previously<sup>40,45</sup> and also detailed in the Supporting Information.

As a calibration, we deposit 80 nm of Au on sapphire substrates and measure  $dR/dT$  at a pump–probe delay time of  $t = 500$  ps using a range of different pump powers, the results of which are shown in Figure 2. The value for  $dR/dT$  that we measure on the 80 nm Au/sapphire calibration for our range of fluences has a small pump power dependence, decreasing from  $dR/dT = 3.74 \times 10^{-5} \text{ K}^{-1}$  to  $dR/dT = 3.2 \times 10^{-5} \text{ K}^{-1}$ , which agrees well with previous measurements for  $dR/dT$  of gold reported by Wang et al.<sup>10</sup> and Tessier et al.<sup>34</sup> This power dependence in  $dR/dT$  is expected on the basis of small perturbations in lattice temperature and the stipulation that  $T_e \approx T_p$ . We also measure this equilibrium  $dR/dT$  at  $t = 500$  ps in our 16 nm Au/sapphire films at various pump powers. As seen in Figure 2, the minor negative dependence on power at this pump–probe delay time remains, but the values for  $dR/dT$  are slightly higher than the 80 nm case ( $dR/dT$  spans  $dR/dT = 4.13 \times 10^{-5} \text{ K}^{-1}$  to  $dR/dT = 3.57 \times 10^{-5} \text{ K}^{-1}$  for these 16 nm films from 5 to 30 mW of incident pump power, respectively).

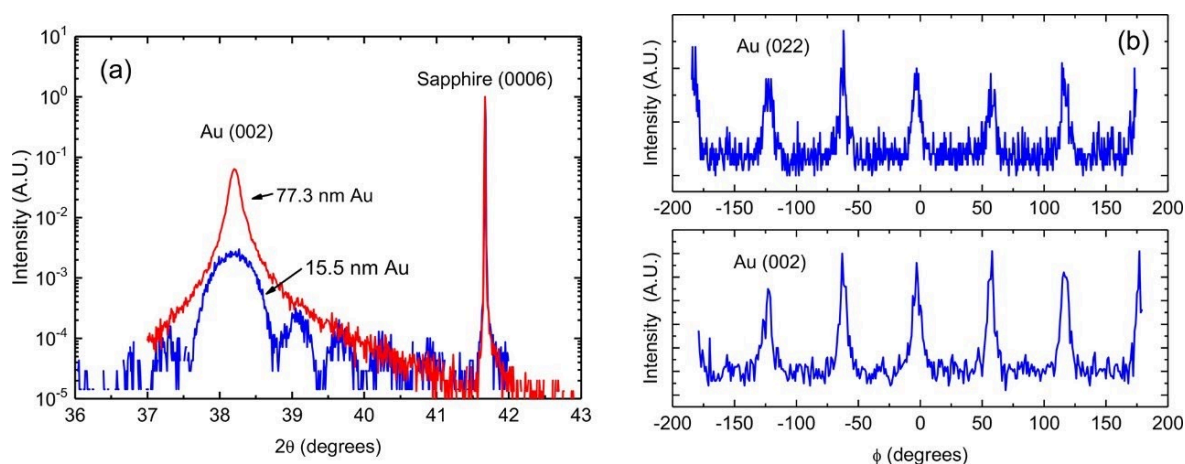


**Figure 2.**  $dR/dT_e$  and  $dR/dT$  measurements as a function of incident pump power for 16 and 80 nm of Au films on sapphire and  $\text{SiO}_2$  substrates.  $dR/dT_e$  in the 16 nm films increase with increasing pump power density during our TDTR experiments. The value for  $dR/dT$  we measure on the 80 nm Au/sapphire calibration decreases from  $dR/dT \approx 4.13 \times 10^{-5} \text{ K}^{-1}$  to  $dR/dT = 3.2 \times 10^{-5} \text{ K}^{-1}$  at 30 mW incident pump power; these values agree well with previous measurements for  $dR/dT$  of gold reported by Wang et al.<sup>10</sup> (solid dashed line) and Tessier et al.<sup>34</sup> (solid dash-dot line). Note, we also measure a negative change in  $dR/dT$  as a function of pump power for the 16 nm Au/sapphire sample in equilibrium conditions ( $T_e \approx T_p$ ) but a value that is slightly higher than both our 80 nm sample and the previous literature values.<sup>10,34</sup> This slight difference in  $dR/dT$  of these Au/sapphire calibration measurements most likely arises from the structural differences between the two films in this study, as demonstrated from our XRD and XRR analysis in Figure 3. The thin gold film has a lower density, which may indicate a higher grain boundary density, smaller crystallite sizes, strain, or other lattice defects. These defects, in turn, may influence electron–phonon coupling to produce a finite thickness-dependent  $dR/dT$  coefficient.<sup>55</sup>

To investigate the origin of this slight difference in  $dR/dT$  of these Au/sapphire films, we characterized the structural properties of these 16 and 80 nm Au films using X-ray diffraction (XRD) and X-ray reflectivity (XRR).  $\theta$ – $2\theta$  scans measure constructive interference of X-rays in a

crystal's planes, by which we can back out information about the material's crystalline structure. By measuring  $\theta$ - $2\theta$  scans, we can determine crystallinity, grain size distribution, and strain of the films, while  $\phi$ -scans tell us the in-plane crystallinity.

X-ray diffraction and reflectivity patterns were collected using a PANalytical Empyrean XRD in parallel beam geometry with a Ge (220) double-bounce hybrid mirror-monochromator ( $1/32^\circ$  divergence slit) incident optic and a  $0.18^\circ$  parallel plate collimator and proportional detector (Xe) as receiving optics. The XRD data in Figure 3a,b show that the films are highly crystalline with an epitaxial (111) orientation on c-plane sapphire. Thickness fringes in the  $\theta$ - $2\theta$  scans (Figure 3a) indicate intense scattering at the Au/substrate interface and smooth interfaces. The RMS roughness and density of the 80 nm film are  $2.3 \pm 0.04$  nm and  $19.8 \pm 0.4$  g cm $^{-3}$ , respectively, and those of the 16 nm film are  $0.8 \pm 0.03$  nm and  $19.1 \pm 0.2$  g cm $^{-3}$ , respectively, as measured by XRR. Note, the thicknesses of these films are also confirmed with XRR, and for the 16 nm film, picosecond acoustics.<sup>53,56</sup> More information on the XRD and XRR methods employed in this work is detailed in the Supporting Information.



**Figure 3.** (a) X-ray diffraction  $\theta$ - $2\theta$  scans of thin (blue line) and thick (red line) Au thin film samples on c-plane sapphire showing, in part, the epitaxial relationship between Au (111) planes and the basal planes of sapphire. The thickness fringes in both scans indicate smooth interfaces in the films. A slight shift of the Au (111) peak in the 15.5 nm thick sample to a lower  $2\theta$  value ( $38.192^\circ$  vs  $38.206^\circ$ ) suggests that compressive biaxial strain may be present. The intensities are normalized by the sapphire (0006) reflection. (b) X-ray diffraction skew-symmetric  $\phi$  scans about the Au (022) and Au (002) peaks (80 nm thick Au sample). The distinct 6-fold symmetry in both scans results from in-plane crystallographic orientation and indicates epitaxial growth of (111)-oriented Au on c-plane sapphire. Note that, for perfect epitaxy, 3-fold symmetry would be expected. The 6-fold symmetry here can be attributed to two equivalent in-plane orientations.<sup>57</sup>

This slight difference in  $dR/dT$  of these Au/sapphire calibration measurements most likely arises from the structural differences between the two films in this study. The thin gold film has a lower density, which may indicate a higher grain boundary density, smaller crystallite sizes, strain, or other lattice defects. These defects, in turn, may influence electron-phonon coupling to produce a finite thickness-dependent  $dR/dT$  coefficient.<sup>55</sup> Our XRD results, as shown in Figure 3a, do suggest biaxial compressive strain could be present in the thin Au films due to a slight shift in the Au (111) peak to lower  $2\theta$ , relative to the 80 nm Au film; an increase in  $dR/dT$  associated with this strain would be consistent with previous TDTR experiments on Al films where strain was determined to cause a change in the thermorefectance coefficient.<sup>10</sup> A comprehensive structural analysis of Au films grown on sapphire is beyond the scope of this work, but our results imply that a material's thermorefectance response depends, at least slightly, on its structural and defect

properties and that these properties can vary from sample to sample. Regardless of the underlying cause, there appears to be a consistent increase in  $dR/dT$  with decreasing film thickness for the values shown in Figure 2.

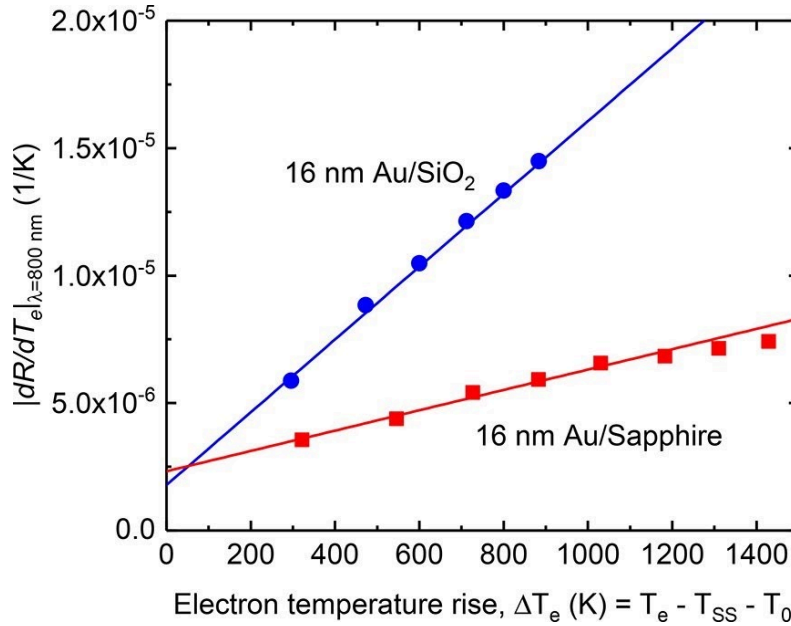
Using eq 7 and replacing  $\Delta T(t)$  with  $\Delta T_e(t = 0)$  calculated with the two temperature model (TTM),<sup>58,59</sup> where  $t = 0$  ps is the time delay corresponding to the peak TDTR signal and maximum electron temperature during electron–phonon nonequilibrium, we calculate  $dR/dT_e$  for the hot electrons in the Au thin film samples; details of these TTM calculations are outlined in the Supporting Information. The results of these  $dR/dT_e$  measurements are shown in Figure 2 as a function of incident pump power. The fact that  $dR/dT_e$  depends on the incident pump power density during a TDTR experiment implies  $dR/dT_e$  exhibits a much stronger temperature dependence than  $dR/dT$ . Thus, this suggests that the hot electron reflectivity changes nonlinearly with temperature (i.e.,  $\Delta R$  is nonlinear with electron temperature), unlike the “equilibrium” reflectivity of a metal. This is consistent with our prior discussion in section “Theory of Temperature Dependence of  $dR/dT_e$ ” and a comparison of both  $dR/dT_e$  and  $dR/dT$  in Figure 2.

## ■ ANALYSIS OF THE ELECTRON TEMPERATURE DEPENDENCE OF $dR/dT_e$

The theory outlined in section “Theory of Temperature Dependence of  $dR/dT_e$ ” suggests that the thermorefectance coefficient for a given material, when grown on varying substrates, could be perceived to be different if the steady-state temperature rises from laser heating are not correctly accounted for. As this “background” temperature rise associated with pulse accumulation is highly dependent on the thermal conductivity of the supporting substrate, the calculated instantaneous temperature rise, and thus  $dR/dT_e$ , can appear to be substrate dependent rather than intrinsic to the thin film material, even when measured under similar laser parameters (e.g., fluence).<sup>45</sup> Indeed, the different values for  $dR/dT_e$  as a function of incident laser power presented in Figure 2 allude to this fact as the nominal value increases with increasing pump power, in contrast to the trend observed under equilibrium conditions. Through the ability to measure  $dR/dT_e$  as a function of  $T_e$  during conditions of strong electron–phonon nonequilibrium (i.e., when  $T_e \gg T_p$ ), we ensure that the electron temperature dependence of  $dR/dT_e$  should always appear linear if our theory outlined in section “Theory of Temperature Dependence of  $dR/dT_e$ ” is correct. Differences in  $dR/dT_e$  of the thin Au films based on the substrate could arise due to different steady-state temperature rises, since the thermal conductivity of sapphire is roughly a factor of 25 greater than  $\text{SiO}_2$ . When  $T_{p,\text{SiO}_2} > T_{p,\text{sapphire}}$ , the expression in parentheses in eq 4 will be different for the sapphire and  $\text{SiO}_2$  substrates. However, since  $T_e \gg T_p$ , this will result in  $dR/dT_e$  varying linearly with  $T_e$  but just having different slopes based on the different values for  $T_p$  resulting from the different substrate thermal conductivities.

To study this, we plot the electron temperature dependence of  $dR/dT_e$  in Figure 4, where the electron temperature rise plotted is obtained via TTM<sup>58,59</sup> calculations. Clearly,  $dR/dT_e$  depends on electron temperature, further demonstrating the nonlinearity of the reflectivity,  $\Delta R$ , during electron–phonon nonequilibrium. Also shown are best fit trend lines to each data set,  $dR/dT_e = C_1 T_e$  where  $C_1$  is a constant, demonstrating that  $dR/dT_e$  is linear with electron temperature in each sample. While the electron temperatures in Au/ $\text{SiO}_2$  and Au/sapphire are predicted to be relatively similar via TTM calculations, we ascribe the difference in  $dR/dT_e$  to be due to the different absolute temperatures of the samples during TDTR measurements, which arise from changes in

steady-state heating based on the different substrates.<sup>45</sup> This is consistent with our discussion in the beginning of this section: namely, the  $dR/dT_e$  is always linear with  $T_e$  assuming  $T_e \gg T_p$ , and changes in the slope of  $dR/dT_e$  vs  $T_e$  can be due to changes in  $T_p$ .



**Figure 4.** Hot electron thermoreflectance coefficient,  $dR/dT_e$ , of the Au/sapphire (squares) and Au/SiO<sub>2</sub> (circles) samples plotted as a function of electron temperature rise,  $\Delta T_e$ , predicted via calculations of the TTM.<sup>58,59</sup>  $dR/dT_e$  of the gold films are linear with  $T_e$  assuming  $T_e \gg T_p$ , regardless of the substrate, and changes in the slope of  $dR/dT_e$  vs  $T_e$  are indicative of changes in  $T_p$ .

## ■ INELASTIC ELECTRON–PHONON SCATTERING RATES DERIVED FROM $dR/dT$

The thermoreflectance coefficients of metals are intimately related to the electron scattering processes through the scattering rate,  $1/\tau$ , in eq 3. As we show here, by examining the coefficients of the expansion of eq 3, the coefficient that describes electron–phonon scattering rates,  $B_{ep}$ , can be simply determined if  $dR/dT$  is known; we outline this approach below. However, we caution that our analysis relies on the metal exhibiting a Drude-like free electron response. Nevertheless, in principle, the extraction of electronic scattering coefficients from measurements of  $dR/dT$  is achievable even for materials that do not obey the Drude model provided an alternative model for the optical dielectric function can be accurately determined.

Consider the thermoreflectance coefficient that describes the temperature response of the reflectivity when the electron and phonon subsystems can be defined with nearly equivalent temperatures (i.e.,  $T_e \approx T_p$ ). In this case, assuming moderate temperatures, the scattering rate term in the Drude model (eq 3) becomes dominated by the electron–phonon scattering rate (i.e.,  $A_{ee}T_e^2 \ll B_{ep}T_p$ ) and the Drude model becomes

$$\tilde{\epsilon}_f(T = T_e \approx T_p) = 1 - \frac{\omega_{\text{plasma}}^2}{\omega(\omega + iB_{ep}T)} \quad (8)$$

where  $T = T_e \approx T_p$ . Using this expression in eq 2 and taking the temperature derivative with respect to  $T$  and then expanding about  $T$  yields

$$\frac{dR(T = T_e \approx T_p)}{dT} \approx \sqrt{R}(\tilde{a} + \tilde{b}T) \quad (9)$$

where  $\tilde{a}$  and  $\tilde{b}$  are complex numbers that are functions of  $\omega$ ,  $\omega_{\text{plasma}}$ , and  $B_{\text{ep}}$ . Thus, in a small perturbation limit where the electron–phonon subsystems are in equilibrium (e.g., the regime of which thermoreflectance spectroscopy techniques such as TDTR are used to measure thermal properties), we can use the linear change in  $dR/dT$  to determine these complex numbers. In the cases where the absolute temperature,  $T$ , is not perturbed enough to obtain this linear relationship and  $dR/dT$  was assumed to be a constant (e.g., the values reported from previous works as shown in Figure 2), one can assume  $dR/dT \approx \tilde{a}\sqrt{R}$ , where

$$\tilde{a} \approx - \frac{2iB_{\text{ep}} \omega_{\text{plasma}}^2}{\omega^3 \sqrt{1 - \frac{\omega_{\text{plasma}}^2}{\omega^2}} \left( 1 + \sqrt{1 - \frac{\omega_{\text{plasma}}^2}{\omega^2}} \right)^2} \quad (10)$$

Equation 10 leads to two major observations. First,  $dR/dT$  in a metal can be expressed solely as a function of  $\omega$ ,  $\omega_{\text{plasma}}$ , and  $B_{\text{ep}}$ . Second, with knowledge of  $dR/dT$ , the electron–phonon scattering rates can be calculated. This offers a potentially alternative way to determine electron–phonon scattering rates in metals that does not rely on interpretation of electrical resistivity or thermal conductivity measurements. Furthermore, since  $dR/dT$  can be measured using continuous wave lasers and without inducing electron–phonon nonequilibrium, this potentially alleviates the necessity of ultrafast laser pump–probe experiments to determine electron–phonon scattering.

Using eq 10 with  $dR/dT$  measured on our 80 nm Au/sapphire and 16 nm Au/sapphire samples, we determine  $B_{\text{ep}} = 2.34 \pm 0.26 \times 10^{11} \text{ s}^{-1} \text{ K}^{-1}$  and  $B_{\text{ep}} = 2.94 \pm 0.22 \times 10^{11} \text{ s}^{-1} \text{ K}^{-1}$ , respectively. Using this same approach applied to the previous measurements of  $dR/dT$  from Wang et al.<sup>10</sup> and Tessier et al.<sup>34</sup> yields  $B_{\text{ep}} = 2.03 \times 10^{11} \text{ s}^{-1} \text{ K}^{-1}$  and  $B_{\text{ep}} = 1.35 \times 10^{11} \text{ s}^{-1} \text{ K}^{-1}$ , respectively. The minor differences among these various values of  $B_{\text{ep}}$  in gold are directly related to our discussion regarding the differences in the measured  $dR/dT$  discussed previously. However, we note that these  $dR/dT$ -derived values for  $B_{\text{ep}}$  are in excellent agreement with values reported for  $B_{\text{ep}}$  in Au previously that were inferred using a range of different techniques other than this thermoreflectance approach presented here ( $B_{\text{ep}} = 1.23\text{--}3.60 \times 10^{11} \text{ s}^{-1} \text{ K}^{-1}$ ).<sup>47,49,52,60</sup> Additionally, we can directly fit eq 8 to our measured change in reflectivity as a function of absolute temperature as a means of confirming these values; in doing so, we find  $B_{\text{ep}} = 2.11 \times 10^{11} \text{ s}^{-1} \text{ K}^{-1}$  for the 16 nm Au film on sapphire, which is in excellent agreement with this alternative approach of using eq 10.

We recently reported elastic electron–phonon dephasing times in Au calculated with time-dependent density functional theory and nonadiabatic molecular dynamics.<sup>61</sup> We explicitly studied the role of the excited states of the electrons on this dephasing time in an effort to assess the role of electron temperature dependence on electron–phonon coupling. On the basis of the degree of electronic excitation, elastic dephasing times spanned from 4.3 to 7.5 fs for a room temperature lattice, which corresponds to  $B_{\text{ep}} = 4.44\text{--}7.75 \times 10^{11} \text{ s}^{-1} \text{ K}^{-1}$ . Note, these rates are faster than  $B_{\text{ep}}$  in gold, as discussed above, but it is important to note our previous work predicted elastic dephasing times, where the experimental determination of  $B_{\text{ep}}$  is related to inelastic electron–phonon processes. The approach to characterize  $B_{\text{ep}}$  via analysis of the  $dR/dT$  value outlined in this work offers a unique path forward to characterize these elastic electron–phonon interactions through control of the sample temperature or geometry to vary the relative roles of elastic and inelastic electron–phonon scattering.<sup>62–64</sup>

## ■ SUMMARY

We have developed an experimental analysis to measure the “hot electron” thermorefectance coefficient,  $dR/dT_e$ , in metal films via monitoring the thermorefectance during pump–probe measurements immediately after pulsed laser absorption. Immediately after pulse absorption during TDTR measurements of thin gold films, the electrons in the Au are out of equilibrium with the lattice, and thus, the condition  $T_e \approx T_p$ , usually assumed while reporting  $dR/dT$  values, is invalid. In this pump–probe time delay measurement regime, which is on the order of the subpicosecond pulse duration,  $T_e$  can be much greater than  $T_p$ , allowing us to measure this hot electron thermorefectance coefficient,  $dR/dT_e$ . Using this approach applied to thin gold films on silica glass and sapphire substrates as experimental examples, we experimentally validate the linear electron temperature dependence of the thermorefectance coefficient. Furthermore, we demonstrate a lattice temperature dependence in  $dR/dT_e$ , suggesting the nonlinear nature of  $\Delta R$  in free electron metals can also arise in part from lattice heating in addition to electron heating. Our approach outlined in this work to directly quantify  $dR/dT_e$  provides a metrology to further understand excited-state scattering effects on the dielectric function of metals.

## ■ ASSOCIATED CONTENT

**Supporting Information.** The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acsp Photonics.8b01045. Two-temperature model; determination of lattice temperature rise via heat diffusion; picosecond acoustics; pulse width and cross correlation of pulses; XRD and XRR experimental method; complex coefficients of eq 4 in the main manuscript (PDF)

## ■ AUTHOR INFORMATION

### Corresponding Author

\*E-mail: phopkins@virginia.edu.

### ORCID

Elizabeth L. Radue: 0000-0003-3377-7126

Ashutosh Giri: 0000-0002-8899-4964

Xin Zhou: 0000-0002-4385-8379

Oleg V. Prezhdo: 0000-0002-5140-7500

Evan L. Runnerstrom: 0000-0002-9054-708X

Patrick E. Hopkins: 0000-0002-3403-743X

### Author Contributions

<sup>‡</sup> E. L. Radue and J. A. Tomko contributed equally to this work.

### Notes

The authors declare no competing financial interest.

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